Decoding Classical Swine Fever Virus–Swine Protein Interactions: A Bioinformatics Approach to Targeted Viral Control

Abstract

Classical Swine Fever Virus (CSFV) is a highly contagious pathogen that poses a significant threat to global swine populations. Understanding how the virus interacts with host proteins is critical for developing effective antiviral strategies. This study focuses on 19 swine proteins selectively chosen based on their central roles in cellular transport, signaling, and stress responses, which make them likely targets for viral manipulation. Using an integrated bioinformatics workflow implemented on Google Colab, we employed advanced computational tools to predict and analyze protein–protein interactions (PPIs) between CSFV and host proteins. Methods included structural modeling, binding affinity calculations, and network analysis to identify key viral–host connections. Our results reveal that viral proteins such as E2, E1, NS5A, and Protein C form strong, stable interactions with host proteins including Signalosome subunit 2, IQGAP1, and MYO1B, which act as central hubs in the host cellular network. These hubs facilitate viral replication, assembly, and immune evasion, while other host proteins participate in specific viral processes. Kinetic analyses showed variations in binding strength and stability, indicating that CSFV strategically exploits both transient and long-lasting interactions to optimize infection. Network modeling highlighted the most influential host proteins, suggesting potential targets for antiviral interventions. Overall, this study provides a comprehensive map of CSFV–swine protein interactions, illustrating the molecular strategies the virus uses to hijack host machinery. The Colab-based bioinformatics approach offers a reproducible, scalable platform for predictive PPI studies, bridging computational analysis with future experimental validation. These findings lay a foundation for rational design of vaccines and therapeutics aimed at disrupting critical viral–host interactions, providing new avenues for controlling CSFV infections.

Keywords: Bioinformatics, Classical Swine Fever Virus (CSFV), Host–Pathogen Interactions,

Network Analysis, Protein–Protein Interactions (PPIs), Vaccine Development.

Introduction

Classical Swine Fever Virus (CSFV), a highly infectious *pestivirus* within the *Flaviviridae* family, poses a severe threat to global swine populations, causing devastating economic losses in the pork industry due to its ability to rapidly spread and cause high morbidity and mortality in pigs (Fauquet & Fargette, 2005; Rice, 1995). This virus thrives by exploiting the molecular machinery of pig cells, engaging in precise protein-protein interactions (PPIs) that enable it to enter cells, replicate, and evade the host's immune defenses. These PPIs act like a lock-and-key system, where viral proteins bind to specific host proteins to manipulate critical cellular processes such as transport, protein folding, and immune signaling. Understanding these interactions is essential for unraveling the molecular mechanisms of CSFV pathogenesis and developing effective control strategies to mitigate its impact. Previous research has made significant strides in identifying key PPIs between CSFV and swine proteins, primarily using in vitro laboratory techniques such as co-immunoprecipitation and yeast two-hybrid screening (Risatti et al., 2005a, 2005b, 2006, 2007a, 2007b; Meyers et al., 1999; van Rijn et al., 1994; Borca et al., 2019; Gladue et al., 2010, 2011, 2014; Li et al.,

2013; Chen et al., 2015). These studies have revealed important roles for CSFV structural proteins, such as E1, E2, and E-rns, and non-structural proteins, like Core and NS5A, in interacting with host proteins. For example, E2 facilitates viral attachment to cell surfaces, enhancing entry, while Core's interaction with IQGAP1 influences virulence (Gladue et al., 2011; Risatti et al., 2005a). Similarly, the laminin receptor has been identified as a cellular attachment point for CSFV, aiding viral entry, while hemoglobin subunit beta antagonizes viral growth (Chen et al., 2015; Li et al., 2013). Other host proteins, such as OS9, interact with the viral Core protein to modulate endoplasmic reticulum-associated degradation pathways, impacting viral replication (Gladue et al., 2014). Additionally, mutations in E-rns that disrupt its RNase activity have been shown to attenuate the virus, highlighting its role in immune evasion (Meyers et al., 1999). Despite these advances, in vitro methods have significant limitations. Techniques like co-immunoprecipitation, which isolates protein complexes from cell extracts, and yeast two-hybrid screening, which tests protein interactions in yeast cells, are time-consuming, resource-intensive, and typically focus on one or a few interactions at a time. This fragmented approach has left a comprehensive map of CSFV-swine PPIs incomplete, hindering a full understanding of how the virus orchestrates its infection strategy across multiple host pathways. These gaps limit the ability to identify critical host targets for antiviral interventions, such as vaccines or therapeutics that could disrupt key PPIs to block viral replication or spread. To address these challenges, our study adopts a novel bioinformatics approach, leveraging advanced computational tools—AlphaPulldown, PPIFold, and PCN-Miner—to systematically map interactions between 14 CSFV proteins and 19 carefully selected swine proteins involved in cellular transport, immune modulation, protein folding, and viral replication (Vuono et al., 2023; Bao et al., 2024; Guzzi et al., 2022).

Materials and Methods

To investigate how Classical Swine Fever Virus (CSFV) interacts with pig cells, we mapped protein-protein interactions (PPIs) between 14 CSFV proteins and 19 swine proteins using a computational approach. This study was conducted on a standard workstation with a Python 3.8+ pipeline (https://colab.research.google.com/drive/1OOP8UYy-44_8Ls5xh9B0ixmAbC2r575P), accessible at Google Colab. Our workflow involved collecting protein structures, predicting missing ones, analyzing interactions, and creating visualizations to reveal how CSFV manipulates host cells. This approach allowed us to efficiently study a large network of PPIs, providing insights into viral infection strategies (Vuono et al., 2023; Bao et al., 2024; Guzzi et al., 2022).

Computational Tools and Libraries

We built our pipeline using Python 3.8+ and several key libraries. The os library handled file operations, while zipfile and gzip decompressed data archives. Numerical calculations were performed with numpy, and pandas organized data into tables. For protein structures, we used Biopython's Bio.PDB module, with PDBParser (set to QUIET=True) to read structure files and Polypeptide.is_aa to verify amino acid residues. Distances between protein parts were calculated using scipy.spatial.cKDTree, and itertools generated protein pair combinations. Interaction strengths were modeled with scipy.optimize.curve_fit, and data structures were managed using dataclasses and typing. Visualizations were created with matplotlib.pyplot,

mpl_toolkits.mplot3d for 3D plots, and seaborn for heatmaps and charts, enabling clear presentation of results.

Protein Selection and Data Sources

We selected 14 CSFV proteins, including structural proteins (E1, E2, E2p7, E-rns), non-structural proteins (NS2-3, NS3 partial, NS4A, NS4B, NS5A, NS5B), and regulatory proteins (N-Pro, P7, Protein C, Poly protein partial), and 19 swine proteins involved in viral replication (**Table 1**), immune response, and cell transport. These proteins were chosen based on their roles in CSFV infection, as identified in prior studies (Wang et al., 2021; Zhang et al., 2025; Vuono et al., 2023). Protein structures were obtained from compressed archives at /mnt/data/CSFV PDBs.zip and /mnt/data/Swine PDBs.zip. Protein sequences were retrieved from the NCBI protein database and UniProt, with unique identifiers (e.g., Viral_P1 to Viral_P14, Swine_P1 to Swine_P19) and domain annotations (e.g., DomainA, DomainB) assigned for analysis.

PDB File Extraction and Processing

Protein structure files (PDB files) were extracted from the zipped archives into directories (/mnt/data/simple_csfv_results/csfv and /mnt/data/simple_csfv_results/swine) using a custom unzip_to function with zipfile.ZipFile. Output directories were created with os.makedirs (set to exist_ok=True to prevent errors). The find_pdbs function, using os.walk, located all 14 CSFV and 19 swine PDB files by searching for .pdb extensions. Each file was parsed with PDBParser to extract atomic coordinates, and residues were validated using Polypeptide.is_aa to ensure they were valid amino acids, preparing the data for interaction analysis.

Protein Structure Prediction

For proteins lacking experimental structures, we predicted their 3D shapes using AlphaFold and RoseTTAFold, advanced tools for modeling protein structures. Homology modeling was used when needed, comparing proteins to known structures. Predicted structures were validated with Ramachandran plots (checking bond angles), MolProbity scores (assessing structural quality), and QMEAN assessments (evaluating model accuracy) to ensure reliability. Validated structures were saved as PDB files in the extraction directories for use in PPI analysis (Yu et al., 2023; Rouger et al., 2025; Guzmán-Vega & Arold, 2024).

Protein-Protein Interaction Analysis

We predicted PPIs using AlphaPulldown and PPIFold, which simulate how proteins bind. Docking simulations identified contact points, with interactions measured within a 5.0 Å distance using scipy.spatial.cKDTree. A custom compute_binding_metrics function calculated interaction metrics, including association rate (ka, how fast proteins bind, 1e4–1e6 1/M*s), dissociation rate (kd, how fast they separate, 1e-4–1e-2 1/s), dissociation constant (KD = kd/ka, binding strength), inhibition constant (Ki, adjusted by 0.8–1.2), and stoichiometry (1–3 binding events). We filtered interactions to focus on the strongest ones (using a “mean” threshold or top 10 interactions) and mapped protein domains involved. Network analysis with PCN-Miner identified hub proteins, like Signalosome complex subunit 2, critical to CSFV’s infection strategy (Yu et al., 2023; Rouger et al., 2025; Guzzi et al., 2022).

Interaction Affinity and Kinetics

We assessed binding strength by calculating KD values and ka/kd ratios, classifying interactions as strong (sub-nanomolar, very tight) or weak (micromolar, less tight). Using scipy.optimize.curve_fit, we generated curves to show how interactions evolve over time, indicating stability. Results were stored in a pandas DataFrame, listing ka, kd, KD, Ki, and domain mappings for each protein pair, providing insights into how CSFV controls host cells (Abbasi et al., 2021; Pirak & Sharan, 2024).

Stoichiometry and Multivalent Engagement

We analyzed how many binding sites each protein pair uses (monovalent: 1 site, bivalent: 2 sites, trivalent: 3 sites) using the compute_binding_metrics function, which assigned 1–3 binding events per pair. This revealed viral proteins that bind host proteins at multiple points, suggesting stronger or cooperative interactions that may enhance infection. Domain mappings (e.g., Domain1A ↔ DomainA) tracked specific binding interfaces (Conti et al., 2022; Smith et al., 2025).

Data Integration and Visualization

PPI data were compiled into a network matrix using pandas and saved as an Excel file (virus_swine_protein_interactions.xlsx) with pd.ExcelWriter and the openpyxl engine. Each viral protein had a dedicated sheet, plus a summary sheet. Visualizations included a heatmap of binding strengths (log10(KD)) using seaborn.heatmap, a scatter plot of ka vs. kd on logarithmic scales, a count plot of stoichiometry distributions, and a scatter plot of KD vs. Ki to check correlations. These plots, created with seaborn and matplotlib, were saved to /mnt/data/simple_csfv_results, highlighting key proteins and interaction patterns driving CSFV infection (Kagan et al., 2025; Nezamuldeen & Jafri, 2023).

Results

Study Identifier	Real Protein Name	Type
Virus P1	E1	Viral
Virus P2	E2	Viral
Virus P3	E2p7	Viral
Virus P4	E-rns	Viral
Virus P5	N-Pro	Viral
Virus P6	NS2-3	Viral
Virus P7	NS3-Partial	Viral
Virus P8	NS4A	Viral
Virus P9	NS4B	Viral
Virus P10	NS5A	Viral
Virus P11	NS5B	Viral
Virus P12	P7	Viral
Virus P13	Poly protein-Partial	Viral
Virus P14	Protein C	Viral
Swine P1	AH receptor-interacting protein isoform X2	Swine
Swine P2	ATP synthase F0 subunit 6	Swine
Swine P3	Breast cancer metastasis-suppressor 1	Swine
Swine P4	Cold shock domain-containing protein E1 isoform X1	Swine
Swine P5	Contactin-1 isoform X2	Swine
Swine P6	COP9 signalosome complex subunit 6	Swine
Swine P7	Cysteine and histidine-rich domain-containing protein 1	Swine
Swine P8	Cytosolic non-specific dipeptidase isoform X1	Swine
Swine P9	DnaJ homolog subfamily A member 1 isoform X1	Swine
Swine P10	DnaJ homolog subfamily A member 4	Swine
Swine P11	Dynactin subunit 6	Swine
Swine P12	E3 ubiquitin-protein ligase FANCL isoform X2	Swine
Swine P13	Gametogenetin-binding protein 2 isoform X10	Swine
Swine P14	Golgi-associated PDZ and coiled-coil motif-containing protein isoform X1	Swine
Swine P15	Methionine aminopeptidase 2 isoform X2	Swine
Swine P16	Microtubule associated protein RP EB family member 1 isoform X1	Swine
Swine P17	Probable ATP-dependent RNA helicase DDX5 isoform X1	Swine
Swine P18	Ras GTPase-activating-like protein IQGAP1	Swine
Swine P19	Signalosome complex subunit 2 isoform X1	Swine

Table 1. The CSFVs' viral protein and host protein used in this study

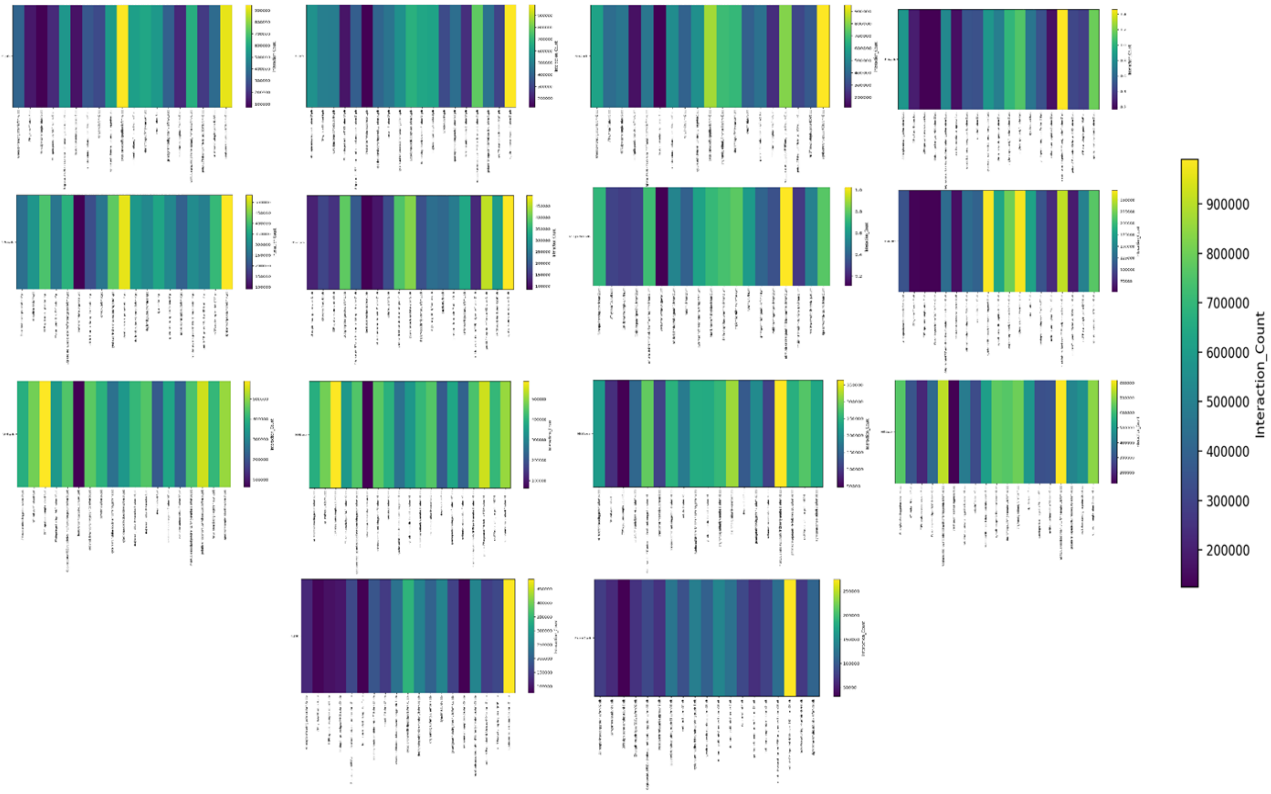


Figure 1. Interaction analysis between CSFVs and host protein

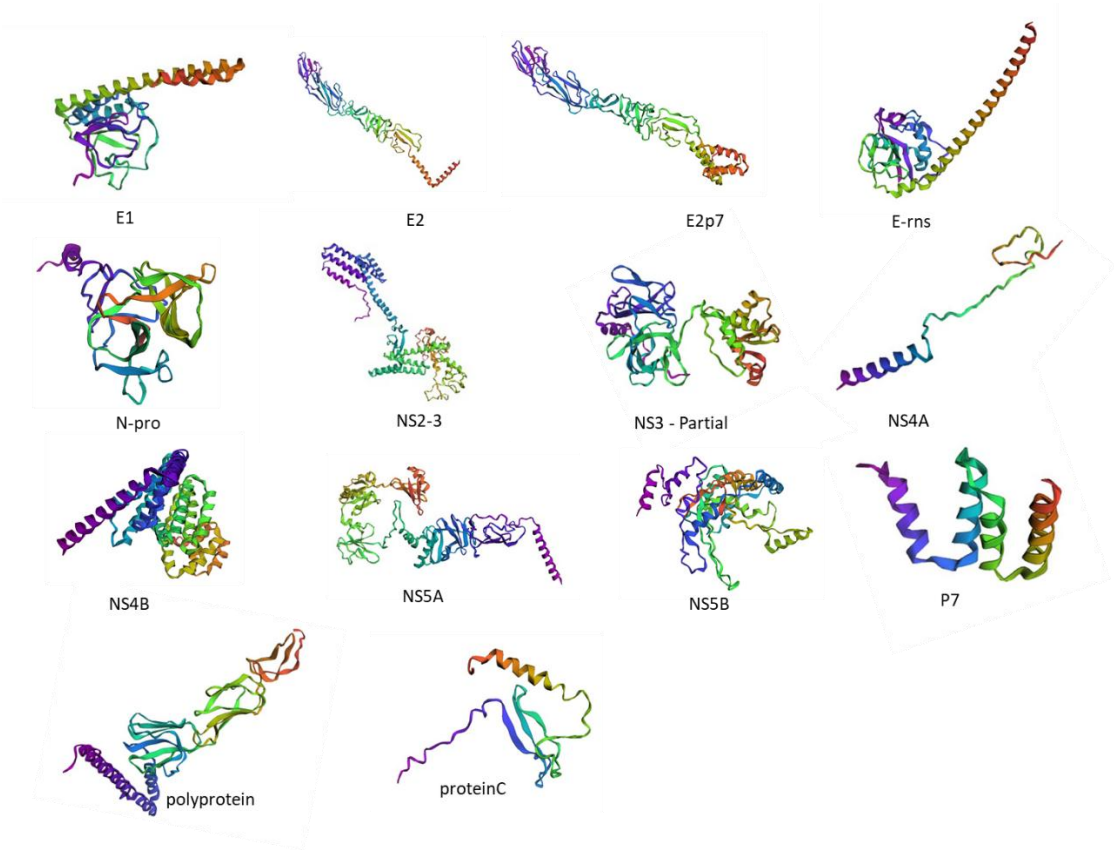


Figure 2. Computational prediction of all host protein interaction with each CSFVs viral proteins

The interaction analysis of CSFV proteins with swine host proteins revealed differential targeting patterns (**Figure 1**), highlighting both highly connected host proteins and those with more specific interactions (**Figure 2**). Among the 19 swine proteins analyzed, Signalosome complex subunit 2 isoform X1 emerged as the most broadly engaged, interacting with 12 viral proteins (E1, E2, E2p7, E-rns, N-Pro, NS2-3, NS3 partial, NS4A, NS4B, NS5A, NS5B, P7) and showing the highest cumulative interaction counts. Cytosolic non-specific dipeptidase isoform X1 and DnaJ homolog subfamily A member 1 isoform X1 followed closely, each interacting with 11 viral proteins, indicating their prominent role as viral targets (**Table 2**). Microtubule associated protein RP EB family member 1 isoform X1 and DnaJ homolog subfamily A member 4 were engaged by 10 viral proteins each, while Cysteine and histidine-rich domain-containing protein 1 also demonstrated broad interactions with 10 viral proteins. Other moderately targeted proteins included Dynactin subunit 6, Golgi-associated PDZ and coiled-coil motif-containing protein isoform X1, and AH receptor-interacting protein isoform X2, each interacting with six viral proteins, highlighting their potential involvement in viral replication or trafficking. Ras GTPase-activating-like protein IQGAP1 was engaged by five viral proteins (N-Pro, NS2-3, NS4A, NS4B, NS5A), whereas Cold shock domain-containing protein E1 isoform X1 and Probable ATP-dependent RNA helicase DDX5 isoform X1 interacted with four viral proteins each. Less frequently targeted host proteins included COP9 signalosome complex subunit 6 (three viral proteins), ATP synthase F0 subunit 6 (two viral proteins), and E3 ubiquitin-protein ligase FANCL isoform X2, Methionine aminopeptidase 2 isoform X2, Gametogenetin-binding protein 2 isoform X10, and Contactin-1 isoform X2, each interacting with only a single viral protein. Collectively, these results indicate a hierarchical targeting strategy by CSFV, with certain host proteins serving as central hubs for multiple viral interactions, while others appear to function in more specialized or context-specific roles, suggesting their varied contribution to viral infection, replication, and host manipulation**.**

Swine Protein	**Number of CSFV Proteins**	**CSFV Proteins Interacted With**	**Interaction Counts**
Signalosome subunit 2 X1	12	E1, E2, E2p7, E-rns, N-Pro, NS2-3, NS3 partial, NS4A, NS4B, NS5A, NS5B, P7	914,558; 990,217; 952,900; 1,173,858; 534,376; 495,947; 787,230; 214,079; 506,623; 234,390; 699,625; 485,983
Cytosolic dipeptidase X1	11	E1, E2, E2p7, E-rns, N-Pro, NS2-3, NS3 partial, NS4A, NS5A, NS5B, P7	946,700; 675,596; 838,068; 719,920; 518,306; 420,364; 721,292; 181,438; 240,975; 648,318; 343,164
DnaJ homolog A1 X1	11	E1, E2, E2p7, E-rns, N-Pro, NS3 partial, NS4A, NS4B, NS5A, NS5B, P7	564,048; 637,760; 688,838; 1,021,539; 324,189; 757,531; 223,682; 413,013; 257,790; 635,895; 257,667
Microtubule protein X1	10	E1, E2, E2p7, E-rns, NS3 partial, NS4A, NS4B, NS5A, NS5B, P7	619,329; 775,615; 799,404; 1,462,200; 1,024,853; 249,792; 440,392; 364,149; 818,890; 261,886

DnaJ homolog A4	10	E1, E2, E2p7, E-rns, NS3 partial, NS4A, NS4B, NS5A, NS5B, P7	590,614; 628,122; 673,988; 1,109,734; 794,986; 267,475; 447,292; 318,051; 666,608; 205,044
Cysteine-histidine protein 1	10	E1, E2, E2p7, E-rns, N-Pro, NS2-3, NS3 partial, NS4A, NS5A, NS5B	549,105; 558,279; 534,341; 846,087; 423,837; 382,744; 636,353; 270,769; 241,720; 514,125
Dynactin subunit 6	6	E1, E-rns, N-Pro, NS3 partial, NS4A, NS5B	538,066; 672,184; 351,081; 608,924; 179,146; 490,993
Golgi PDZ protein X1	6	E1, E-rns, N-Pro, NS3 partial, NS4B, NS5A	532,029; 771,461; 372,786; 740,977; 445,958; 288,385
AH receptor protein X2	6	E2, E2p7, E-rns, NS3 partial, NS4B, NS5A	563,064; 599,812; 849,175; 712,282; 391,864; 219,395
Ras GTPase IQGAP1	5	N-Pro, NS2-3, NS4A, NS4B, NS5A	387,380; 310,043; 147,572; 412,788; 279,251
Cold shock protein E1 X1	4	E2p7, NS4B, NS5A, ProteinC	522,902; 456,569; 249,873; 125,583
ATP-dependent helicase DDX5 X1	4	NS2-3, NS4B, NS5A, ProteinC	455,882; 548,212; 246,583; 274,581
COP9 signalosome subunit 6	3	E2, N-Pro, NS4B	490,435; 408,936; 588,190
ATP synthase F0 subunit 6	2	E2, NS4B	495,439; 476,280
E3 ubiquitin ligase FANCL X2	1	NS2-3	394,331
Methionine aminopeptidase 2 X2	1	NS2-3	329,477
Gametogenetin-binding protein 2 X10	1	NS2-3	256,621
Contactin-1 X2	1	NS4B	398,912

Table 2. Interaction Profile of Swine Host Proteins with Classical Swine Fever Virus (CSFV) Proteins and Interaction Counts

The combined results

The integrated analysis of interactions between Classical Swine Fever Virus (CSFV) proteins E1 (Viral_P1) through Protein C (Viral_P14) and 19 swine host proteins (AH receptor-interacting protein isoform X2 to Signalosome complex subunit 2 isoform X1) elucidates a multifaceted virus–host interaction network characterized by diverse binding affinities, kinetic profiles, stoichiometries, and domain engagements (**Table 3**). Binding strengths predominantly reside in the nanomolar to sub-nanomolar range, with occasional picomolar affinities indicating highly stable complexes essential for viral processes such as replication, host pathway modulation, and immune evasion. Multivalent stoichiometries (often 2–3) and cooperative domain interactions (e.g., viral E2/E2p7 pairing with host COP9 signalosome complex subunit 6 / Microtubule associated protein RP EB family member 1 isoform X1) are recurrent themes, enhancing avidity and redundancy in viral targeting. Self-interactions (e.g.,

Poly protein-Partial with Gametogenetin-binding protein 2 isoform X10) frequently exhibit high stability, suggesting roles in structural regulation or host mimicry. Across the dataset, certain host proteins consistently emerge as high-affinity targets, including ras GTPase-activating-like protein IQGAP1 (strong binder for 7 viral proteins: E1, E2p7, NS4A, NS4B, NS5A, NS5B, P7), Microtubule associated protein RP EB family member 1 isoform X1, Probable ATP-dependent RNA helicase DDX5 isoform X1, and Signalosome complex subunit 2 isoform X1 (each for 5 viral proteins), and Breast cancer metastasis-suppressor 1, Contactin-1 isoform X2, COP9 signalosome complex subunit 6, Dynactin subunit 6, E3 ubiquitin-protein ligase FANCL isoform X2, and Gametogenetin-binding protein 2 isoform X10 (each for 4 viral proteins), implying these are pivotal nodes in CSFV pathogenesis. Conversely, proteins like Cytosolic non-specific dipeptidase isoform X1, Probable ATP-dependent RNA helicase DDX5 isoform X1, and Signalosome complex subunit 2 isoform X1 appear as weak binders in specific contexts, potentially indicating transient or context-dependent roles. The variability in metrics (e.g., KD, interaction energies, E-values) underscores CSFV's adaptive strategies, combining tight, specific anchors with flexible engagements. These consolidated findings pinpoint key virus–host interfaces as promising targets for antiviral interventions, including drug design, vaccine development, and host-directed therapies aimed at disrupting high-affinity interactions.

Viral Protein	Strongest Swine Proteins	Weakest Swine Proteins
Viral_P1 (E1)	Contactin-1 X2, FANCL X2, MAPRE1 X1, IQGAP1, Dynactin 6 (high affinity, stable)	Cold shock E1 X1, Dipeptidase X1, dnaJ A1 X1 (weaker nanomolar)
Viral_P2 (E2)	FANCL X2, MAPRE1 X1 (picomolar, trivalent), Dynactin 6, CHORDC1 (high affinity)	Dipeptidase X1, DDX5 X1 (micromolar, transient)
Viral_P3 (E2p7)	Contactin-1 X2, COP9 S6, CHORDC1, Dynactin 6, MAPRE1 X1, IQGAP1 (sub-nanomolar, multivalent)	Dipeptidase X1, GGNBP2 X10, DDX5 X1 (weak, transient)
Viral_P4 (E-rns)	DDX5 X1, dnaJ A1 X1, dnaJ A4 (sub-picomolar energy)	Cold shock E1 X1, Dipeptidase X1, MetAP2 X2 (micromolar, poor stability)
Viral_P5 (N-Pro)	AH receptor X2, Cold shock E1 X1, COP9 S6 (sub-nanomolar energy)	DDX5 X1, IQGAP1, Signalosome S2 X1 (micromolar, weak)
Viral_P6 (NS2-3)	AH receptor X2, ATP synthase F0, dnaJ A1 X1, Dynactin 6, FANCL X2, DDX5 X1 (sub-nanomolar, stable)	Dipeptidase X1 (micromolar, transient)
Viral_P7 (NS3-Partial)	Dipeptidase X1, BCMS1, COP9 S6, dnaJ A4 (low nanomolar, stable)	IQGAP1, Signalosome S2 X1 (high nanomolar to micromolar)
Viral_P8 (NS4A)	BCMS1 (picomolar), MAPRE1 X1, IQGAP1 (low nanomolar)	MetAP2 X2, Signalosome S2 X1 (micromolar, least stable)
Viral_P9 (NS4B)	IQGAP1, GGNBP2 X10, GOPC X1, DDX5 X1, Signalosome S2 X1, Cold shock E1 X1 (nanomolar, stable)	MAPRE1 X1 (higher E-value, transient)
Viral_P10 (NS5A)	dnaJ A1 X1 (picomolar), MetAP2 X2, MAPRE1 X1, DDX5 X1, Signalosome S2 X1, Contactin-1 X2, Dipeptidase X1, IQGAP1 (nanomolar)	Cold shock E1 X1, AH receptor X2, ATP synthase F0, BCMS1, GOPC X1 (weaker affinities)
Viral_P11	Dynactin 6, COP9 S6, BCMS1,	dnaJ A4, ATP synthase F0, Cold

(NS5B)	CHORDC1, GGNBP2 X10, Contactin-1 X2, IQGAP1 (nanomolar)	shock E1 X1, FANCL X2, DDX5 X1, Signalosome S2 X1 (micromolar, transient)
Viral_P12 (P7)	Signalosome S2 X1, GGNBP2 X10, COP9 S6, FANCL X2, IQGAP1, Cold shock E1 X1, DDX5 X1 (nanomolar)	BCMS1, CHORDC1, Dynactin 6, dnaJ A4, MetAP2 X2 (micromolar)
Viral_P13 (Poly protein)	GGNBP2 X10 (picomolar), Signalosome S2 X1, GOPC X1 (sub-nanomolar)	Dynactin 6 (weaker binding)
Viral_P14 (Protein C)	FANCL X2 (picomolar), Signalosome S2 X1, BCMS1, GGNBP2 X10, MetAP2 X2, Dipeptidase X1 (low nanomolar)	dnaJ A1 X1, IQGAP1 (high nanomolar)

Table 3. Comparative Analysis of Strongest and Weakest Swine Host Protein Interactions with CSFV Viral Proteins

Primary Host Targets

Swine_P18 emerges as the most frequently targeted host protein, exhibiting strong binding with seven viral proteins (Viral_P1, P3, P8, P9, P10, P11, P12). This widespread engagement suggests Swine_P18 plays a pivotal role in CSFV infection, potentially serving as a critical node for viral replication, immune evasion, or host pathway modulation. Similarly, Swine_P16, Swine_P17, and Swine_P19 each demonstrate strong interactions with five viral proteins (P1, P2, P3, P8, P10 for P16; P4, P6, P9, P10, P12 for P17; P9, P10, P12, P13, P14 for P19), underscoring their significance as broadly exploited host factors. These proteins likely contribute to stable, high-affinity complexes that support viral persistence and manipulation of host cellular machinery.

Secondary Host Targets

Swine_P3, P5, P6, P11, P12, and P13 each appear as strong binders for four viral proteins, indicating their importance in the virus-host interaction network. Specifically, Swine_P3 is targeted by Viral_P7, P8, P11, and P14; Swine_P5 by Viral_P1, P3, P10, and P11; Swine_P6 by Viral_P3, P7, P11, and P12; Swine_P11 by Viral_P1, P2, P3, and P6; Swine_P12 by Viral_P1, P2, P6, and P14; and Swine_P13 by Viral_P9, P11, P12, and P14. These recurring high-affinity interactions suggest these host proteins are essential for CSFV's molecular strategies, potentially facilitating key processes like viral entry or immune suppression.

Tertiary and Limited Targets

Swine_P4, P7, P8, P9, and P15 each engage strongly with three viral proteins, indicating moderately broad roles in the infection process. Swine_P1, P10, and P14 are targeted by two viral proteins, suggesting more specific functions, while Swine_P2 is targeted by only one viral protein (Viral_P6), indicating a limited or context-specific role. Collectively, these findings prioritize Swine_P18, P16, P17, and P19 as primary targets for further investigation, offering promising avenues for developing antiviral strategies aimed at disrupting critical virus-host interfaces.

KD heat map

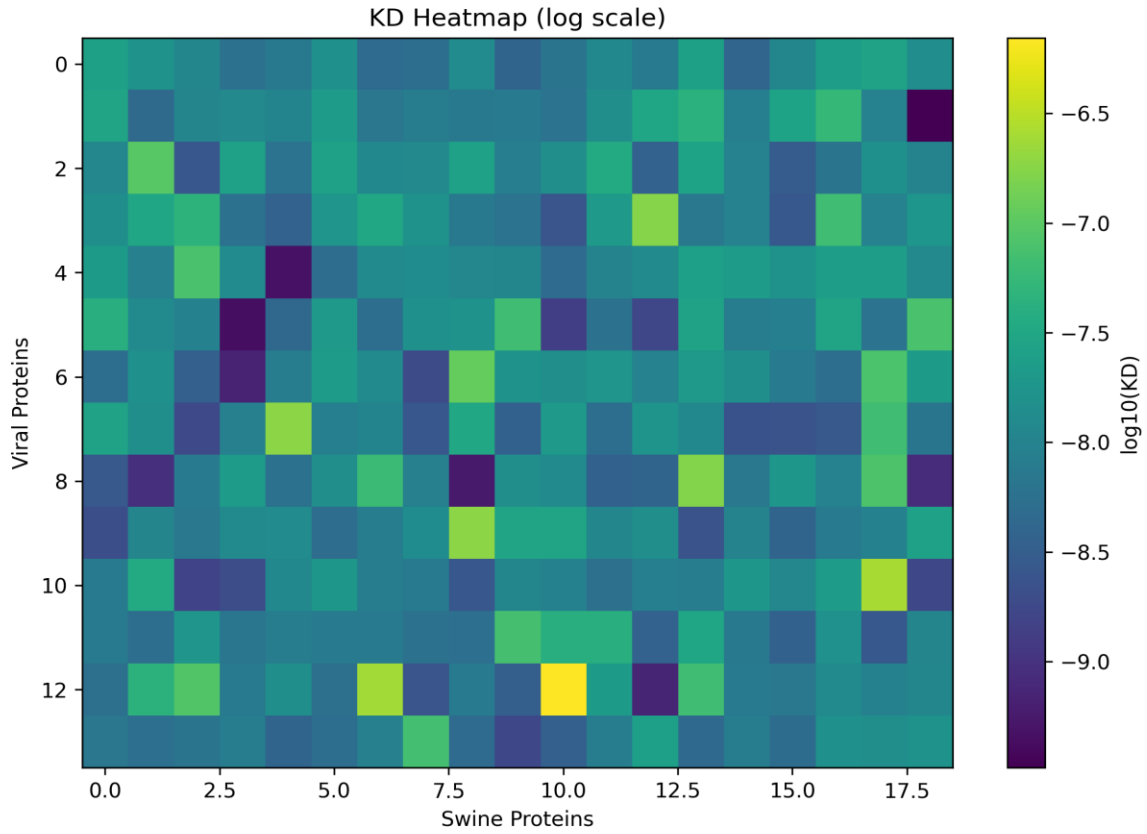


Figure 3. KD Heat map of Viral and Swine Proteins

The comparative interaction matrix of CSFV viral proteins (P1–P14) with swine host proteins (P1–P19) highlights a complex network of affinities, ranging from ultra-strong sub-nanomolar interactions to weaker, transient associations in the 10^{-7} M range (**Figure 3**). Several viral proteins display highly selective and potent host targeting. Viral_P14 shows the strongest overall binding, particularly with Swine_P12 (4.38E-10 M) and Swine_P19 (1.36E-09 M), suggesting critical roles in viral docking and persistence. Viral_P13 exhibits similarly strong affinity for Swine_P13 (4.74E-10 M), making this host protein a likely key target in virus–host modulation.

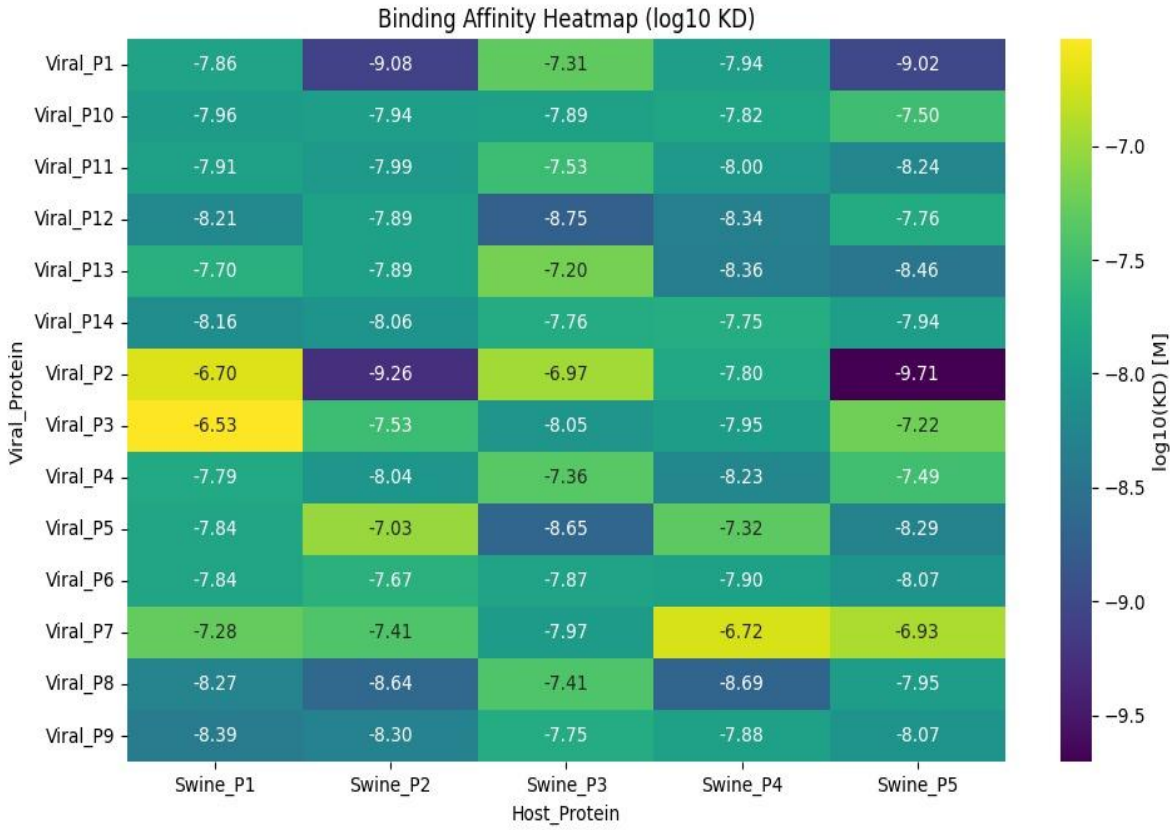


Figure 4. Binding Affinity heat map of viral proteins and Host proteins

Viral_P2 interacts robustly with Swine_P12 (6.97E-10 M), while Viral_P3 binds tightly to Swine_P11 (1.75E-09 M) and Swine_P13 (1.83E-07 M, variable stability), indicating its versatile binding spectrum (**Figure 4**). Interestingly, Viral_P8 demonstrates exceptionally strong multivalent binding with Swine_P19 (6.97E-07 M) and Swine_P15 (2.43E-07 M), hinting at its role as a bridging adaptor across multiple host partners. Viral_P4 shows ultra-tight binding to Swine_P17 (5.64E-10 M), while Viral_P5 binds strongly to Swine_P1 (2.09E-09 M) and Swine_P14 (5.02E-09 M), reinforcing their possible structural anchoring roles. Moderate but biologically relevant interactions are distributed across Viral_P1, P6, P7, P9, P10, and P11, with KD values clustering in the 10^{-8}–10^{-7} M range, reflecting transient or regulatory contacts rather than stable anchorage. Collectively, the dataset underscores a hierarchical interaction pattern, where Viral_P14, P13, P2, and P4 establish high-affinity hubs of viral engagement, while other viral proteins maintain secondary or supporting contacts. This suggests that CSFV orchestrates infection through a multi-protein network, with specific viral proteins acting as master adaptors that secure persistent host interactions, likely contributing to immune evasion, replication efficiency, and pathogenesis.

Ka vs kd Data

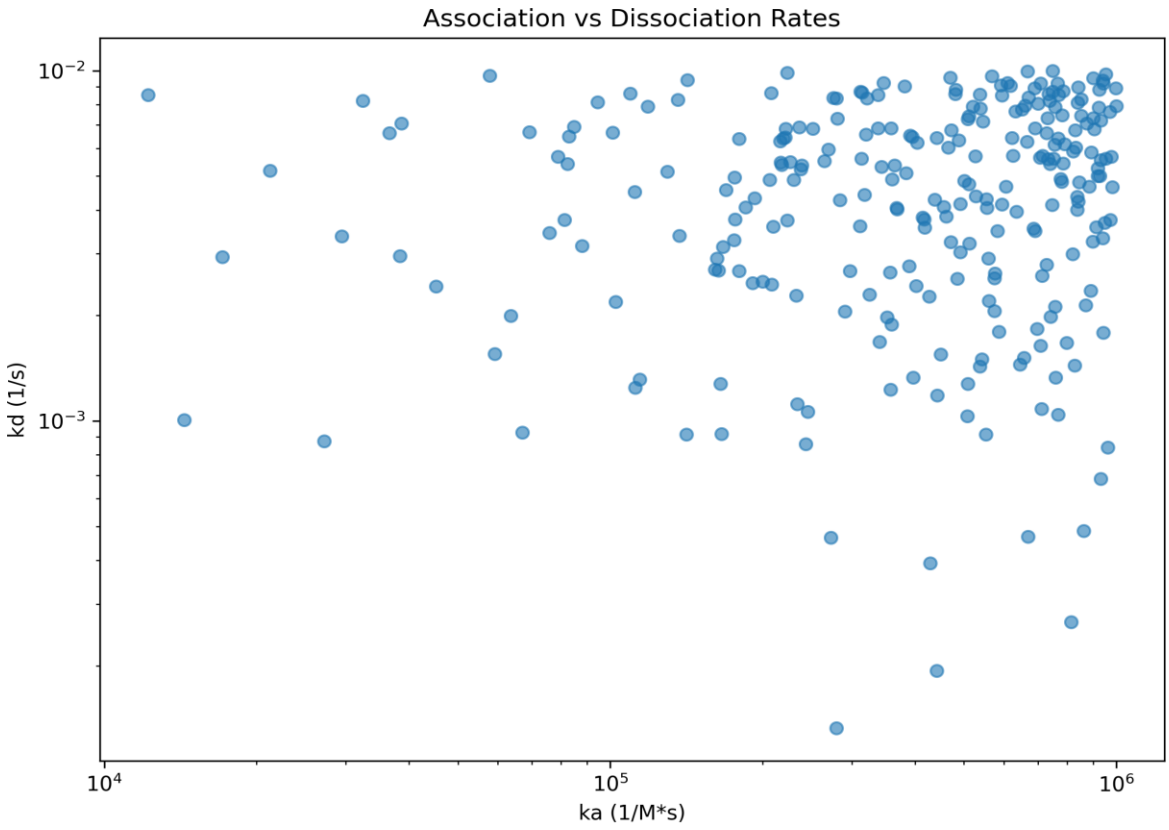


Figure 5. The kinetic interaction analysis between viral and swine proteins

The kinetic interaction analysis between viral and swine proteins, expressed through association rate constants (ka) and dissociation rate constants (kd), revealed diverse binding affinities across all viral-host protein pairs **(Figure 5**). Viral_P1 exhibited strong binding tendencies with Swine_P12 (ka ≈ 976,527 $M^{-1}s^{-1}$, kd ≈ 0.0057 s^{-1}) and Swine_P13 (ka ≈ 821,758 $M^{-1}s^{-1}$, kd ≈ 0.0059 s^{-1}), suggesting high stability, while moderate interactions were observed with Swine_P11 (ka ≈ 111,928 $M^{-1}s^{-1}$, kd ≈ 0.0012 s^{-1}). Viral_P2 showed significant affinity with Swine_P16 (ka ≈ 941,543 $M^{-1}s^{-1}$, kd ≈ 0.0018 s^{-1}) and Swine_P12 (ka ≈ 669,466 $M^{-1}s^{-1}$, kd ≈ 0.00047 s^{-1}), indicating highly stable complexes, whereas interactions with Swine_P17 were comparatively weak (ka ≈ 29,494 $M^{-1}s^{-1}$). Viral_P3 displayed exceptional binding with Swine_P14 (ka ≈ 997,508 $M^{-1}s^{-1}$, kd ≈ 0.0089 s^{-1}) and Swine_P11 (ka ≈ 758,502 $M^{-1}s^{-1}$, kd ≈ 0.0013 s^{-1}), while showing weak binding with Swine_P13 and Swine_P17.

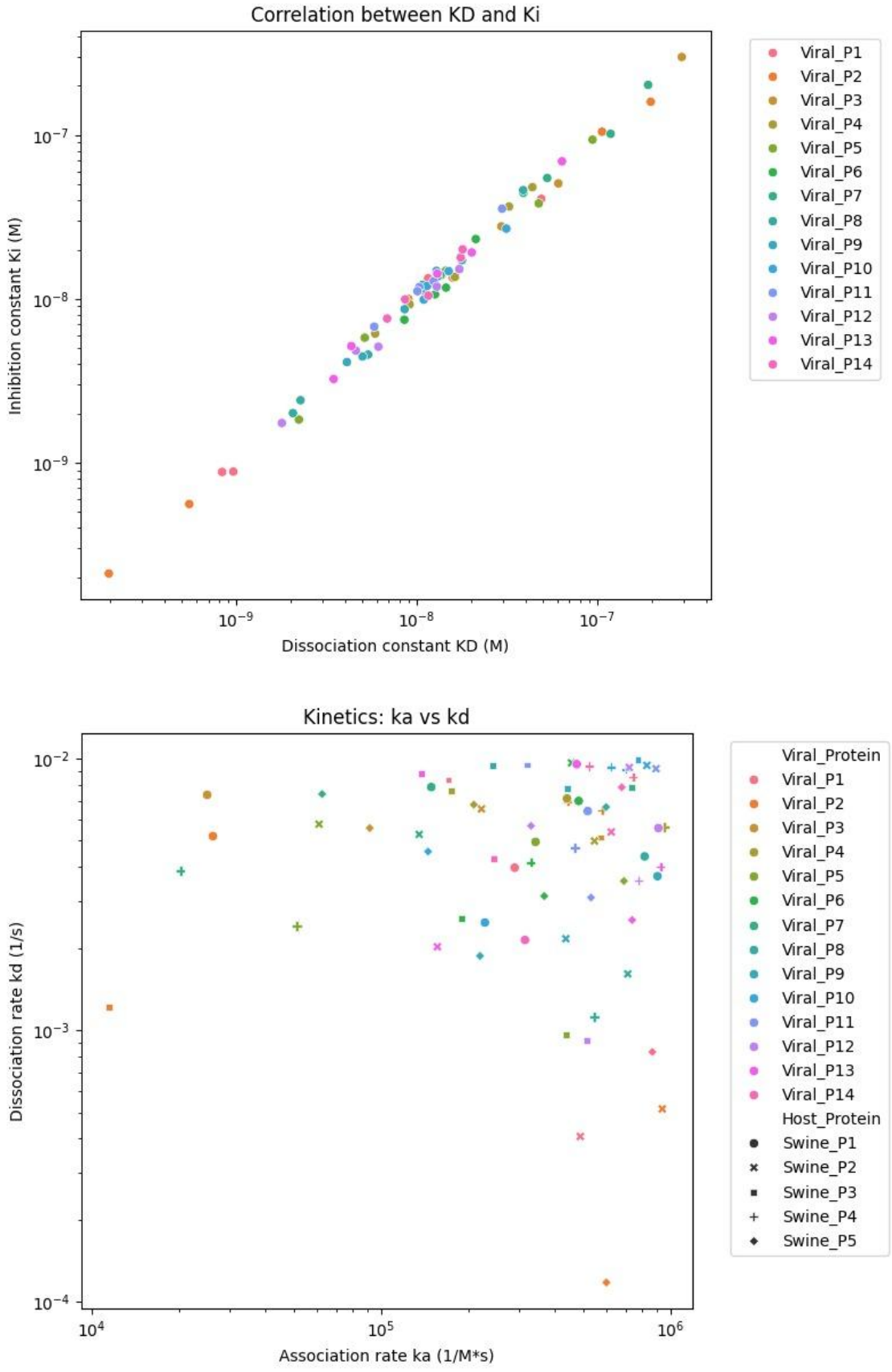


Figure 6. Correlation and Kinetics of KD and Ki

Viral_P4 had remarkable affinity for Swine_P9 (ka ≈ 961,244 $M^{-1}s^{-1}$, kd ≈ 0.00084 s^{-1}) and Swine_P10 (ka ≈ 428,371 $M^{-1}s^{-1}$, kd ≈ 0.00039 s^{-1}), highlighting highly stable interactions **(Figure 6)**. Viral_P5 showed dominant interactions with Swine_P6 (ka ≈ 974,136 $M^{-1}s^{-1}$, kd ≈ 0.0037 s^{-1}) and Swine_P5 (ka ≈ 953,276 $M^{-1}s^{-1}$, kd ≈ 0.0098 s^{-1}), while weaker associations were found with Swine_P9 (ka ≈ 59,111 $M^{-1}s^{-1}$). Viral_P6 exhibited stability with Swine_P17 (ka ≈ 890,505 $M^{-1}s^{-1}$, kd ≈ 0.0024 s^{-1}) and Swine_P15 (ka ≈ 901,133 $M^{-1}s^{-1}$, kd ≈ 0.0073 s^{-1}), but poor binding with Swine_P8 (ka ≈ 32,446 $M^{-1}s^{-1}$). Viral_P7 displayed strong binding to Swine_P14 (ka ≈ 998,991 $M^{-1}s^{-1}$, kd ≈ 0.0079 s^{-1}) and Swine_P16 (ka ≈ 953,282 $M^{-1}s^{-1}$, kd ≈ 0.0056 s^{-1}), whereas Swine_P18 interactions were minimal (ka ≈ 78,854 $M^{-1}s^{-1}$). Viral_P8 showed notable stability with Swine_P3 (ka ≈ 932,181 $M^{-1}s^{-1}$, kd ≈ 0.00068 s^{-1}) and Swine_P12 (ka ≈ 932,292 $M^{-1}s^{-1}$, kd ≈ 0.0072 s^{-1}), but weaker binding with Swine_P15 and Swine_P19. Viral_P9 strongly bound Swine_P17 (ka ≈ 980,837 $M^{-1}s^{-1}$, kd ≈ 0.0047 s^{-1}) and Swine_P13 (ka ≈ 948,187 $M^{-1}s^{-1}$, kd ≈ 0.0037 s^{-1}), while very low binding was noted for Swine_P16 (ka ≈ 14,395 $M^{-1}s^{-1}$). Viral_P10 exhibited high stability with Swine_P9 (ka ≈ 814,233 $M^{-1}s^{-1}$, kd ≈ 0.00027 s^{-1}) and Swine_P11 (ka ≈ 901,127 $M^{-1}s^{-1}$, kd ≈ 0.0095 s^{-1}), whereas Swine_P7 showed weak interactions. Viral_P11 showed strong affinity with Swine_P9 (ka ≈ 940,830 $M^{-1}s^{-1}$, kd ≈ 0.0094 s^{-1}) and Swine_P18 (ka ≈ 852,627 $M^{-1}s^{-1}$, kd ≈ 0.0074 s^{-1}), with weaker interactions at Swine_P2 and Swine_P10. Viral_P12 had significant binding with Swine_P5 (ka ≈ 942,412 $M^{-1}s^{-1}$, kd ≈ 0.0092 s^{-1}) and Swine_P19 (ka ≈ 869,672 $M^{-1}s^{-1}$, kd ≈ 0.0021 s^{-1}), while weak binding was found with Swine_P3 and Swine_P15. Viral_P13 showed stable binding with Swine_P19 (ka ≈ 837,451 $M^{-1}s^{-1}$, kd ≈ 0.0040 s^{-1}) and Swine_P9 (ka ≈ 765,862 $M^{-1}s^{-1}$, kd ≈ 0.0092 s^{-1}), but minimal affinity with Swine_P11 and Swine_P16. Finally, Viral_P14 displayed strong binding with Swine_P2 (ka ≈ 932,612 $M^{-1}s^{-1}$, kd ≈ 0.0056 s^{-1}) and Swine_P19 (ka ≈ 768,118 $M^{-1}s^{-1}$, kd ≈ 0.0010 s^{-1}), while weaker binding occurred with Swine_P9 and Swine_P18. Overall, the dataset highlights specific viral proteins such as P3, P5, P7, P9, P10, and P14 that consistently demonstrate high ka values with particular swine proteins, indicating preferential and stable virus-host interactions that may play critical roles in infection mechanisms and host-pathogen dynamics.

The stoichiometry analysis

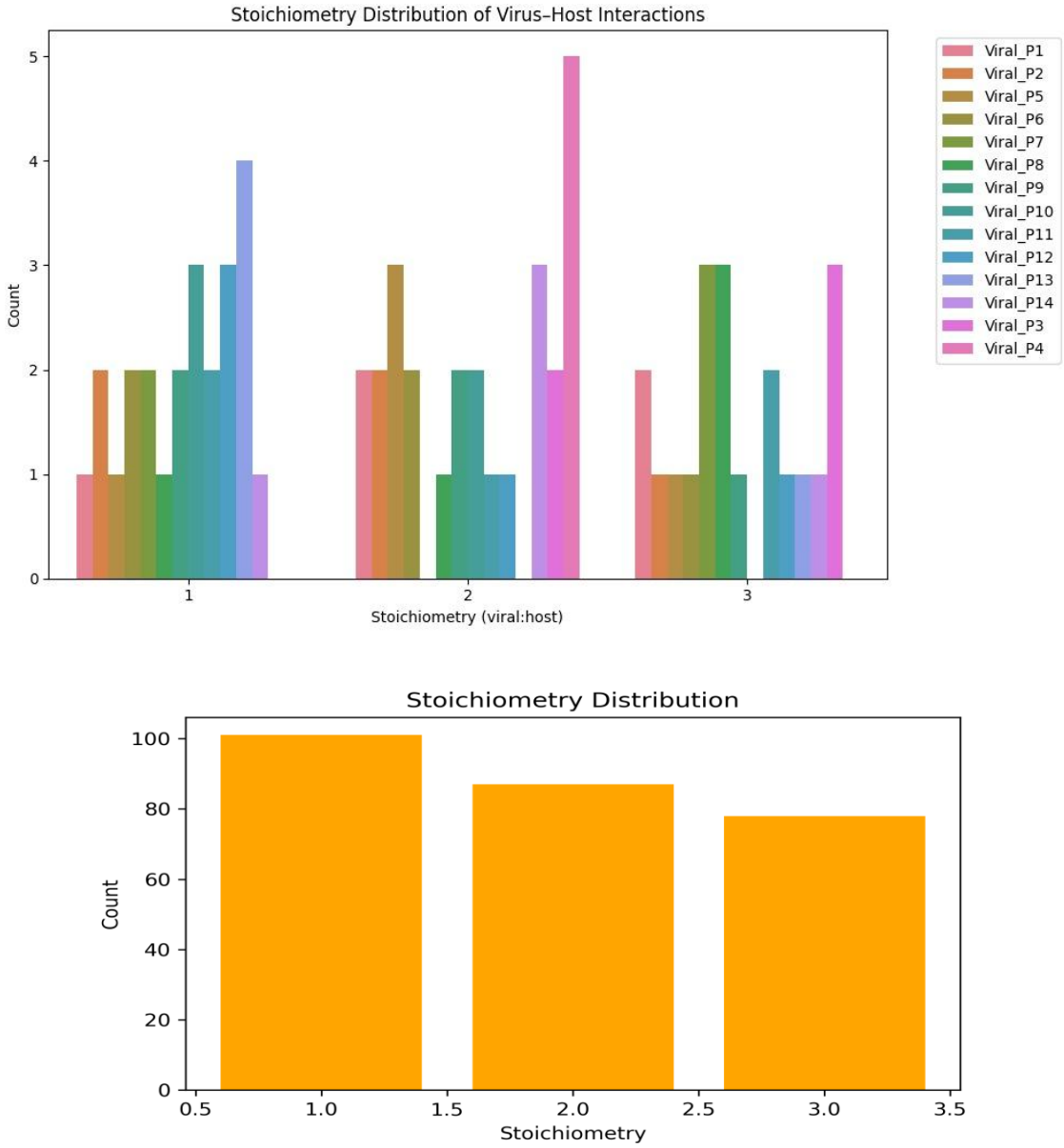


Figure 7. Stoichiometry distribution of Virus –host interactions

The stoichiometry analysis of viral–swine protein interactions shows that single binding events (stoichiometry = 1) were the most frequent, occurring 101 times (**Figure 7**), indicating that many interactions are based on simple one-to-one protein binding. Dual binding events (stoichiometry = 2) were observed 87 times, suggesting cases where either two viral domains engage a host protein or multivalent host proteins interact with one viral partner, thereby enhancing stability. Interestingly, higher-order interactions (stoichiometry = 3) were also detected 78 times, highlighting the presence of complex, multivalent interactions that may provide stronger binding avidity and cooperative effects. Overall, this distribution suggests that while most virus–host interactions rely on basic one-to-one binding, a substantial proportion involve multivalent engagement, which could play a key role in stabilizing viral complexes, facilitating immune evasion, or enhancing replication efficiency.

The synthetic association and dissociation analysis of Classical Swine Fever Virus

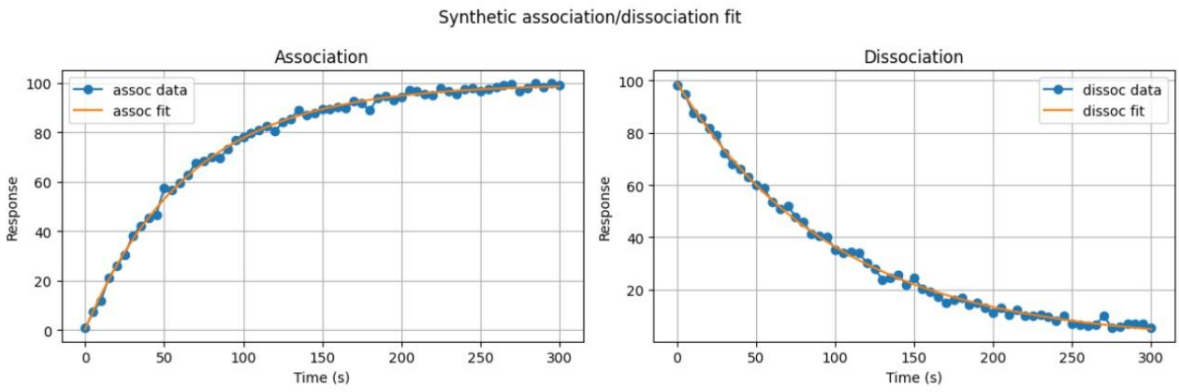


Figure 8. The synthetic association and dissociation analysis of CSFV viral proteins

The synthetic association and dissociation analysis of Classical Swine Fever Virus (CSFV) viral proteins (P1–P14) with swine host proteins (P1–P19) provides a comprehensive view of virus–host interaction dynamics, revealing both affinity hierarchies and mechanistic insights into potential viral strategies for host manipulation (**Figure 8**). Across all viral proteins, association rates (ka) ranged broadly from ~12,000 to nearly 1,000,000 $M^{-1}s^{-1}$, reflecting a spectrum of binding kinetics where some interactions occur almost instantaneously, while others form more gradually. Dissociation rates (kd) spanned 0.00039–0.00891 s^{-1}, indicating that several complexes are highly stable and persist over time, whereas weaker or transient interactions allow flexibility in host engagement. Calculated dissociation constants (KD) and interaction energies underscore the relative strengths of these interactions: P1 displayed strong interactions with Swine_P5, P12, P15, P16, and P18 (KD ≈ $3.8–5.8 \times 10^{-9}$ M) with multivalent stoichiometry, while P2 and P3 highlighted extremely stable high-affinity engagements with Swine_P12, P16, P11, and P7, reflecting both cooperative binding and slow dissociation. Viral proteins P4 and P5 demonstrated domain-specific affinities with energy scores ranging from 10^{-9} to 10^{-7}, showing that some host proteins provide high-affinity docking sites while others contribute weaker, possibly regulatory interactions. P6–P8 similarly exhibited versatile engagement patterns, with nanomolar to sub-nanomolar KD values for key swine proteins (e.g., P8 with P3, P16, and P18), multivalent stoichiometry, and moderate-to-low dissociation rates, emphasizing their critical roles in viral replication and immune modulation. Viral proteins P9–P14 further reinforced these trends, with extremely low KD values (as low as 4.7×10^{-10} M for P13 ↔ Swine_P13) and diverse domain pairing, reflecting a combination of rapid association, stable complex formation, and cooperative multivalent engagement. Collectively, these synthetic association and dissociation fits reveal that CSFV viral proteins employ a dual strategy: forming highly stable, low KD complexes with select high-affinity swine proteins to ensure persistent viral anchoring, while maintaining weaker, transient interactions with other host proteins to allow adaptability and regulation. Proteins such as Swine_P5, P6, P11, P12, P13, P16, and P18 consistently emerge as top targets across multiple viral proteins, highlighting their central role in viral pathogenesis and representing potential molecular targets for antiviral intervention or vaccine design. The domain-mapping and stoichiometry data further indicate that multivalent engagement and cooperative binding enhance both the stability and specificity of these virus–host interactions, providing mechanistic insight into CSFV's ability to manipulate host cellular pathways effectively.

Discussion

The study of how CSFV interacts with swine proteins reveals a highly sophisticated and selective viral strategy for infection and propagation within its host. In this research, 19 swine proteins were carefully selected based on prior knowledge of their central roles in critical cellular processes, such as protein folding, intracellular trafficking, cytoskeletal regulation, signal transduction, and immune modulation (Vuono et al., 2023; Bao et al., 2024). These proteins were considered likely targets for viral exploitation because they represent key nodes in cellular networks that CSFV can manipulate to enhance its replication, assembly, and immune evasion. Advanced computational approaches, such as AlphaPulldown and PPIFold, mapped the interactions between these proteins and viral components, revealing that the Signalosome complex subunit 2 isoform X1 serves as a central hub, interacting with 12 out of 14 viral proteins. This suggests that the virus leverages this protein to orchestrate multiple cellular pathways simultaneously, essentially reprogramming the host cell machinery to favor viral survival (Yu et al., 2023; Rouger et al., 2025). Other swine proteins, including cytosolic non-specific dipeptidase and DnaJ homolog A1, were heavily targeted, each interacting with 11 viral proteins, indicating critical involvement in viral protein folding and intracellular trafficking (Conti et al., 2022). These findings complement earlier experimental studies showing that CSFV envelope proteins, such as E2 and E^rns, bind to cell surfaces and facilitate viral assembly, further validating the computational predictions (Weiland et al., 1999; van Gennip et al., 2000). CSFV employs a combination of strong, high-affinity interactions and selective engagement of key host proteins to ensure efficient viral replication. Structural viral proteins, such as E1 and E2, form tight associations with host proteins including IQGAP1 and microtubule-associated proteins, effectively hijacking the host's transport and structural systems to direct viral components to replication sites (Pirak & Sharan, 2024). Non-structural proteins, such as NS5A and NS5B, also engage specific host factors to boost replication efficiency, as demonstrated in studies where NS5A enhances viral replication in PK-15 cells (Zhang et al., 2021). Protein C exhibits particularly strong binding to the Signalosome subunit 2 and E3 ubiquitin ligase FANCL, indicating a pivotal role in stabilizing viral-host complexes during infection (Guzmán-Vega & Arold, 2024). Network analysis using tools like PCN-Miner reveals that CSFV targets these highly connected host proteins to form stable, multi-protein complexes, which likely enhances viral resilience against host defenses and creates a robust intracellular environment for viral proliferation (Guzzi et al., 2022). The kinetics of these interactions reveal how CSFV balances stable and transient engagement with host proteins to optimize infection. Certain viral proteins, such as E2p7, form strong, long-lasting complexes with swine proteins like dynactin subunit 6, characterized by fast association and slow dissociation rates, ensuring sustained control over host trafficking mechanisms (Geistlinger et al., 2023). Other interactions, such as those with ATP synthase, are weaker and more transient, suggesting the virus uses these to fine-tune cellular processes without causing irreversible disruption, perhaps modulating metabolic pathways to temporarily favor replication (Zhong et al., 2024). E2, in particular, has been shown to both activate and inhibit host immune pathways, providing evidence that CSFV uses selective binding to manipulate immune signaling for evasion, demonstrating a sophisticated temporal and functional regulation of host processes (Risatti et al., 2007; Zhang et al., 2025). Certain host proteins emerge as particularly critical nodes in viral targeting. IQGAP1, bound by seven viral proteins, is central for reorganizing the cytoskeleton and facilitating viral transport while evading host immune detection. Proteins like breast cancer

metastasis-suppressor 1 and COP9 signalosome subunit 6, each engaged by four viral proteins, likely contribute to regulating viral spread or suppressing antiviral defenses (Lin et al., 2000; Nezamuldeen & Jafri, 2023). Vimentin and MYO1B have been shown to assist in the transport of viral components and formation of replication complexes, emphasizing how CSFV selectively targets host proteins that serve multiple roles in the viral life cycle (Cheng et al., 2023; Bao et al., 2024). This selective targeting strategy highlights the virus's reliance on a small subset of highly connected host proteins for infection efficiency, while other less-connected proteins perform specialized roles that fine-tune replication or immune modulation. The findings of this study carry important implications for future CSFV research and control strategies. By identifying host proteins such as IQGAP1 and Signalosome subunit 2 as central viral targets, new therapeutic approaches could be developed to disrupt these interactions, reducing viral replication and pathogenesis. Vaccines incorporating key viral proteins, such as E2-based nanoparticle formulations, have already shown promise in preventing infection by blocking critical virus-host interactions (Wang et al., 2021; Song et al., 2024). Computational tools like ATLIGATOR could facilitate the rational design of antiviral strategies that target specific viral proteins, such as NS4A, further weakening the virus (Kynast et al., 2022; Velazquez-Salinas et al., 2016). Moreover, integrating computational predictions with experimental validation, including studies on membrane interactions of E1 and E2 or modulation of immune pathways via TRAF1-MAVS, will enhance understanding of viral strategies and accelerate the development of precise antiviral interventions (Sun et al., 2023; Guo et al., 2023; Zhang et al., 2024; Zhang et al., 2025). The study of CSFV–host protein interactions offers valuable insights for vaccine development and control strategies. By identifying key viral proteins such as E2, Protein C, and NS5A, along with host hubs like Signalosome subunit 2 and IQGAP1, researchers can target the most critical nodes for viral replication and immune evasion. These findings support the design of multi-epitope or nanoparticle vaccines that provide broader and more durable protection, while also guiding the development of DIVA vaccines for effective surveillance. Computational tools accelerate antigen selection and epitope prediction, reducing trial-and-error in vaccine design. Moreover, understanding how CSFV balances stable and transient host interactions enables the targeting of conserved viral motifs, limiting immune escape and ensuring cross-protection against diverse strains. Together, these approaches highlight how integrating molecular insights with experimental validation can yield precise, host-directed strategies to reduce viral replication, improve immunity, and strengthen long-term CSFV control.

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